Middle ear & mastoid

The middle ear cleft as a six-walled box: what lies on each wall, the ossicles, muscles, nerves, folds and the mastoid you open in surgery.

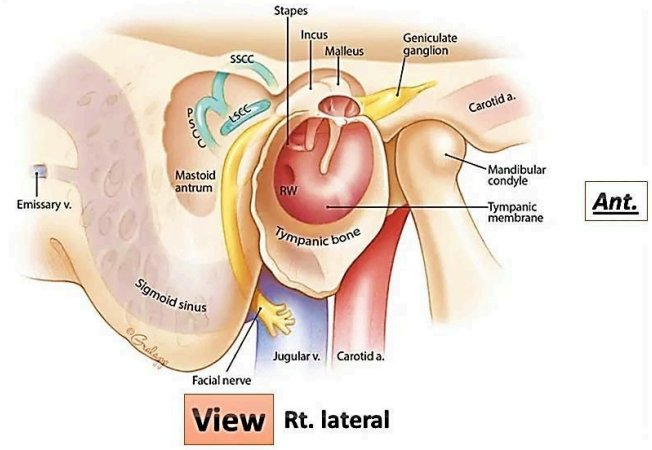
1. Middle ear cleft and its parts
2. The six walls
3. Facial recess and sinus tympani
4. Ossicles
5. Intratympanic muscles
6. Nerves of the middle ear
7. Mucosal folds and compartments
8. Mastoid antrum and air cells
9. Blood supply and lymphatics
10. Vascular and nerve variants in the middle ear
11. In theatre
12. On imaging

01 Middle ear cleft and its parts

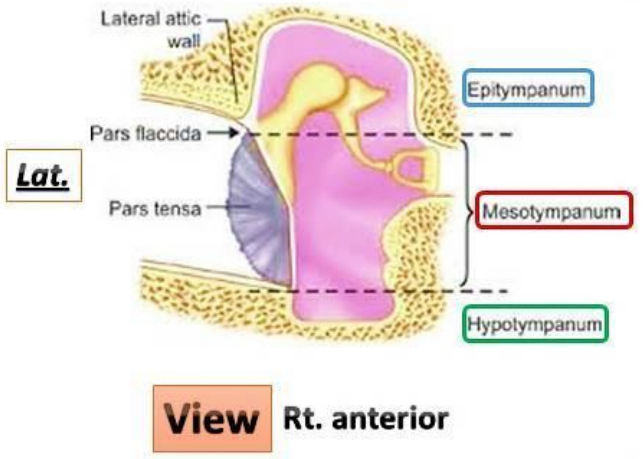
- **Middle ear cleft:** ET, middle ear, aditus, mastoid antrum and mastoid air cells. Lined by mucosa. The tympanic cavity is about 1–2 cm³; the whole cleft is larger and depends on mastoid pneumatisation.
- **Contents:** 3 ossicles, 2 muscles (tensor tympani, stapedius), 2 nerves (chorda tympani, tympanic plexus).
- **Mesotympanum:** at the level of the pars tensa. **Epitympanum (attic):** above it, medial to Shrapnell's membrane and the scutum. **Hypotympanum:** below the pars tensa. **Protympanum:** around the ET opening.

CLINICAL PEARL Neighbours that make complications possible: tegmen above (temporal lobe), jugular bulb below, ICA in front, sigmoid sinus behind, labyrinth medially.

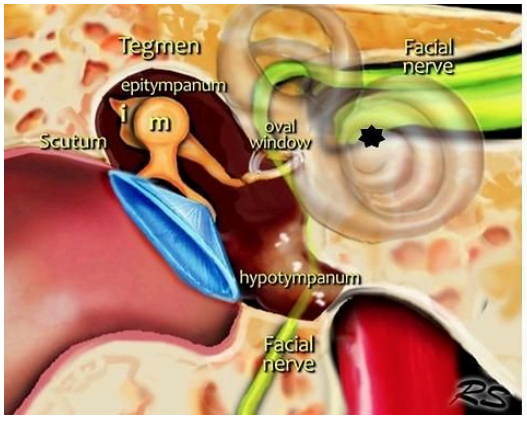
SOURCES DIFFER VOLUME: ENT Steps gives 1–2 cm³ for the whole cleft; standard texts give that figure for the tympanic cavity alone.



Middle ear and its surroundings, right lateral view
ENT Steps Anatomy, p. 40



Epitympanum, mesotympanum and hypotympanum
ENT Steps Anatomy, p. 41



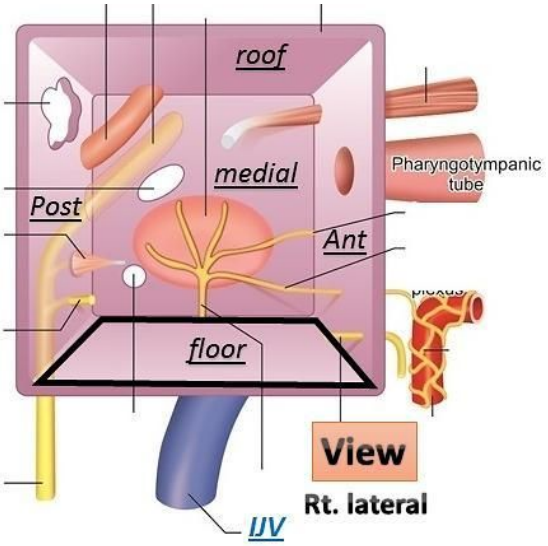
Coronal section through the middle ear

ENT Steps Anatomy, p. 49

02 The six walls

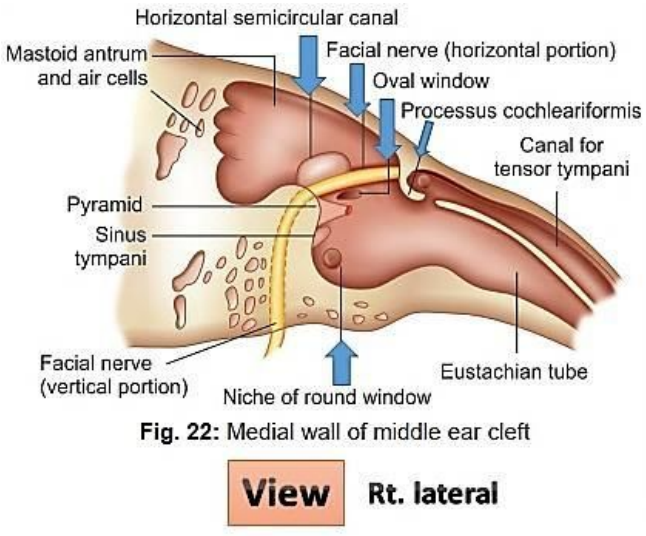
- **Roof (tegmental):** tegmen tympani, continuous back as the tegmen antri; separates from the middle cranial fossa.
- **Floor (jugular):** thin bone over the **jugular bulb**; may be dehiscent.
- **Anterior (carotid):** thin bone over the **ICA**, the **ET** opening, the **canal for tensor tympani** above it, the canal for chorda tympani (anterior canaliculus) and the anterior malleal ligament.
- **Posterior (mastoid): aditus ad antrum** above, **pyramid** (stapedius tendon) below it, vertical facial nerve behind, **facial recess** lateral and **sinus tympani** medial to the pyramid.
- **Medial (labyrinthine): promontory** (basal turn of cochlea), **oval window** (stapes footplate), **round window** (secondary TM), horizontal **facial canal** above the oval window, **lateral SCC** above the facial canal, **processus cochleariformis** in front of the oval window.
- **Lateral (membranous):** TM below, bony **scutum** (lateral attic wall) above.

CLINICAL PEARL A dehiscent jugular bulb can bulge into the middle ear and be injured at myringotomy or cochlear implant surgery. The tympanic facial nerve is dehiscent in many people, which is why acute otitis media can cause facial palsy.



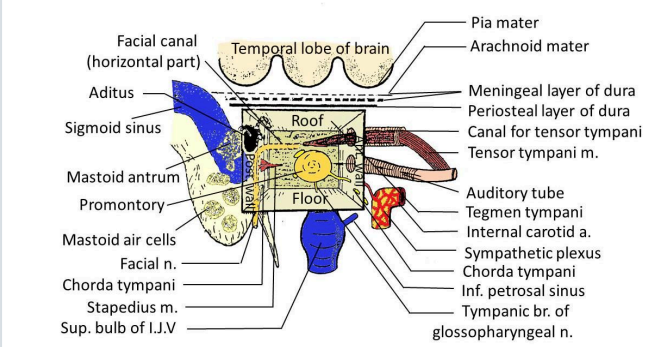
The six walls of the middle ear as a box

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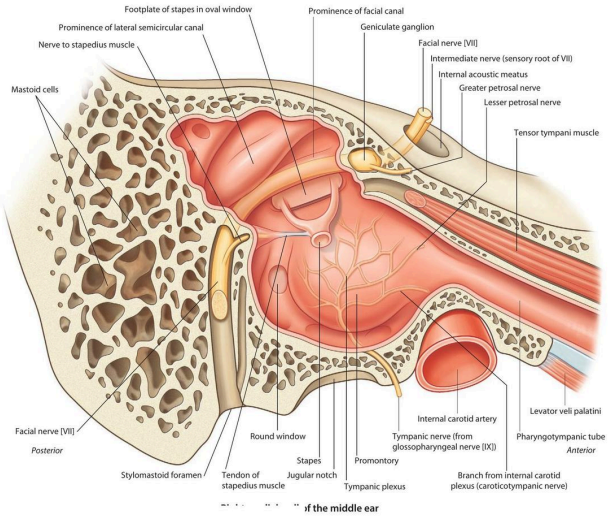
Medial wall of the middle ear cleft

ENT Steps Anatomy, p. 48



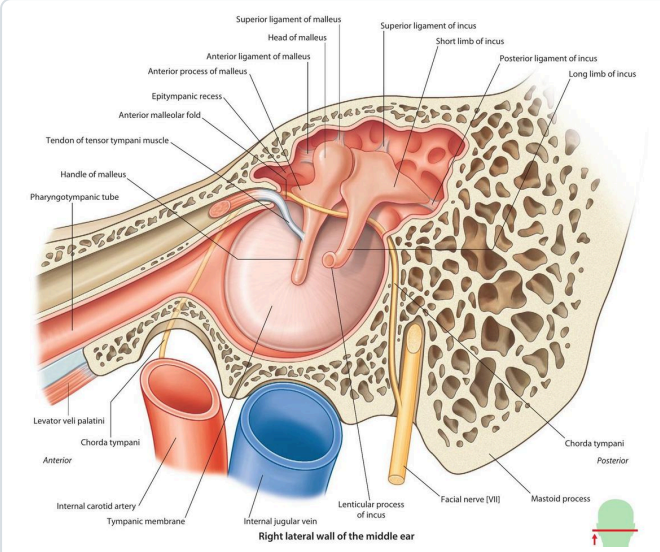
Middle ear walls in coronal section

Dr. Sameh Doss handout, p. 185



Right medial wall of the middle ear

Gray's Atlas of Anatomy, 2nd ed., p. 522



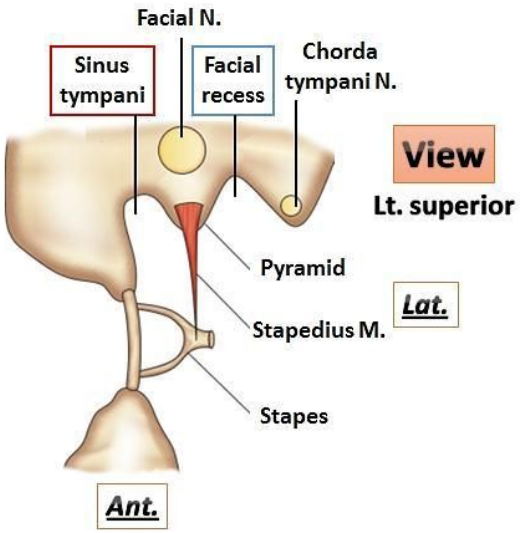
Right lateral wall of the middle ear

Gray's Atlas of Anatomy, 2nd ed., p. 523

03 Facial recess and sinus tympani

- **Facial (suprapyramidal) recess:** lateral to the pyramid. Medial: vertical facial nerve. Lateral: **chorda tympani** and annulus. Above: **fossa incudis** (short process of incus).
- **Sinus tympani (infrapyramidal recess):** medial to the pyramid and the facial nerve, between the ponticulus above and subiculum below.
- **Posterior tympanotomy** opens the middle ear through the facial recess without taking down the canal wall: used in combined-approach tympanoplasty and cochlear implantation.

> **CLINICAL PEARL** The sinus tympani is hidden from both the canal and the mastoid view, so residual cholesteatoma most often hides here.



Posterior wall: facial recess and sinus tympani
ENT Steps Anatomy, p. 46

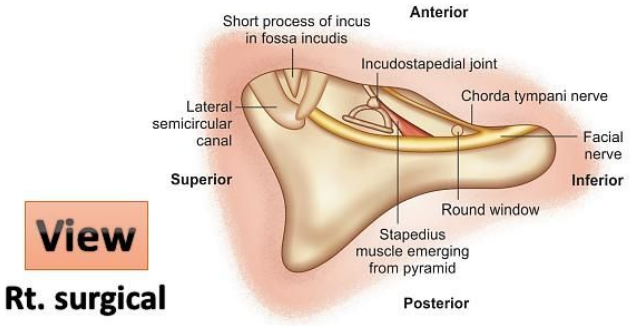


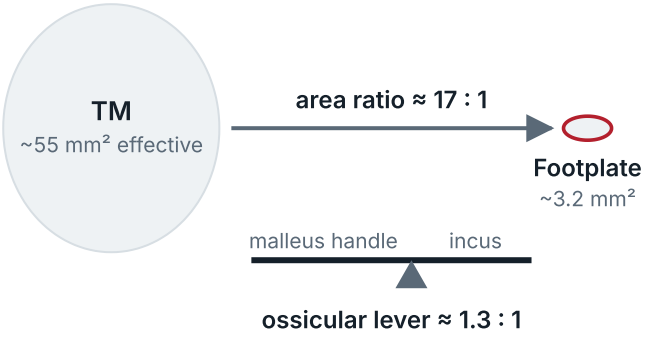
Fig. 20: Posterior tympanotomy. Structures of middle ear seen through the opening of facial recess

Posterior tympanotomy: middle ear seen through the facial recess
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04 Ossicles

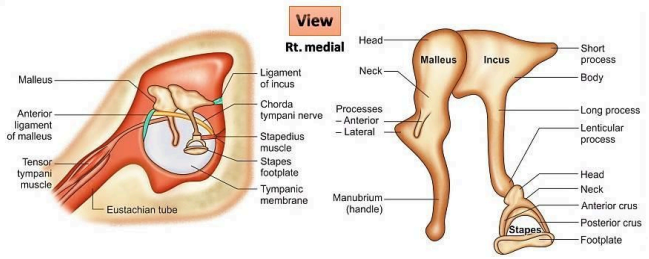
- **Malleus** (8 mm, largest): head and neck in the attic; **handle embedded in the TM**; lateral process seen as a knob on the drum; anterior process.
- **Incus:** body and short process in the attic (short process in the fossa incudis); **long process** hangs down to form the **incudostapedial joint**.
- **Stapes** (3.5 mm, smallest bone in the body): head, neck, anterior and posterior crura, **footplate** held in the oval window by the annular ligament.
- **Impedance matching:** the TM is about 17 times larger than the footplate and the ossicular lever adds about 1.3×, so pressure rises about 22 times (roughly 25–30 dB). Without it most sound would reflect off the fluid-filled cochlea.
- Origin: malleus head and incus body from the 1st arch; handle of malleus, long process of incus and stapes superstructure from the 2nd arch; footplate partly from the otic capsule.

> **CLINICAL PEARL** The long process of incus has the poorest blood supply, so it is the part most often eroded in chronic otitis media.



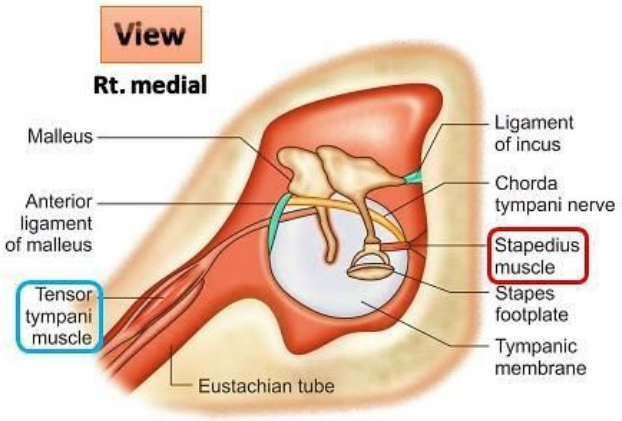
17 × 1.3 ≈ 22× pressure gain ≈ 25–30 dB

Why a total ossicular discontinuity behind an intact drum gives a maximal conductive loss of about 60 dB, while a perforation alone causes much less.



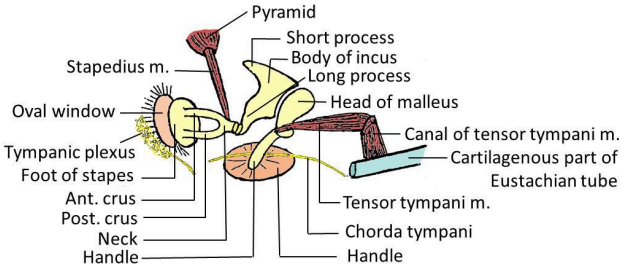
The three ossicles

ENT Steps Anatomy, p. 50



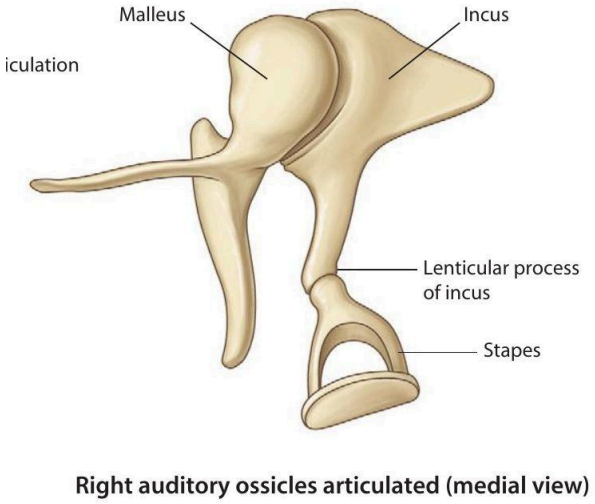
Medial view of TM, ossicles and intratympanic muscles

ENT Steps Anatomy, p. 51



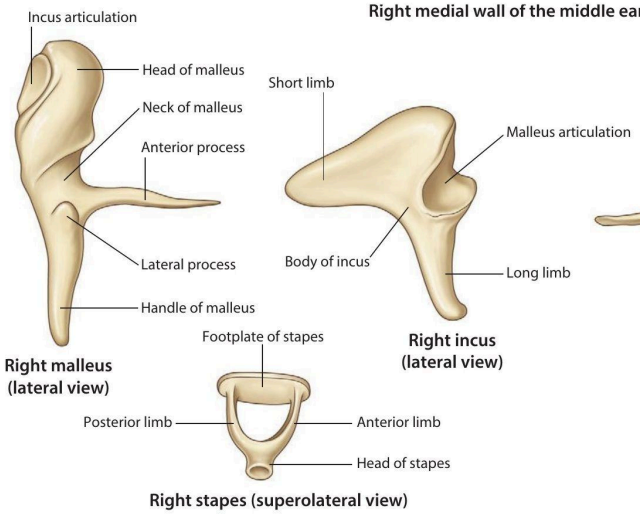
Ossicles and intratympanic muscles

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Right auditory ossicles articulated (medial view)

Gray's Atlas of Anatomy, 2nd ed., p. 522



Right malleus, incus and stapes

Gray's Atlas of Anatomy, 2nd ed., p. 522

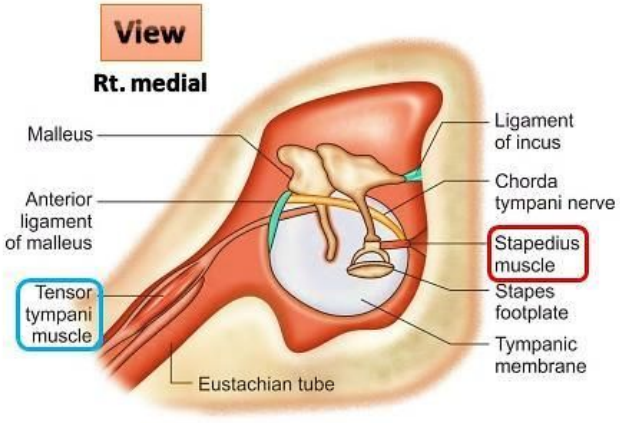
05 Intratympanic muscles

- **Tensor tympani** (1st arch, **V3**): from its canal above the ET; tendon turns round the **processus cochleariformis** to the neck of the malleus. Tenses the TM.

- **Stapedius** (2nd arch, **nerve to stapedius from VII**): from the pyramid to the neck of the stapes. Dampens loud sound.
- **Acoustic (stapedial) reflex:** sound over about 80 dB contracts stapedius; protects the cochlea, works as gain control, and reduces self-generated noise during chewing and speech.

CLINICAL PEARL Hyperacusis in Bell's palsy means the lesion is above the nerve to stapedius.

SOURCES DIFFER TENSOR TYMPANI ORIGIN: ENT Steps: bony canal above the ET. Doss: cartilaginous part of the ET. Gray's lists both plus the greater wing of sphenoid.



Medial view of TM, ossicles and intratympanic muscles
ENT Steps Anatomy, p. 51

06 Nerves of the middle ear

- **Tympanic plexus** on the promontory: Jacobson's nerve (IX) plus caroticotympanic sympathetic fibres. Supplies the medial TM, cavity, mastoid cells and bony ET.
- **Parotid secretomotor pathway:** inferior salivatory nucleus → IX → Jacobson's nerve → tympanic plexus → **lesser petrosal nerve** → **otic ganglion** → auriculotemporal nerve → parotid.
- **Chorda tympani (VII):** enters through the posterior canaliculus, runs on the medial surface of the TM **between the malleus handle and the long process of incus**, leaves through the petrotympanic fissure. Carries **taste from the anterior two-thirds of the tongue** and secretomotor fibres to the submandibular and sublingual glands.

CLINICAL PEARL Referred otalgia: pain from the tonsil or tongue base reaches the ear through IX (Jacobson's nerve) and from the larynx and hypopharynx through X (Arnold's nerve).

SOURCES DIFFER CHORDA TYMPANI ON THE TM: ENT Steps: between the malleus and long process of incus. Doss: between the handle of malleus and the mucosa. Both are true: lateral to the incus, medial to the malleus handle.

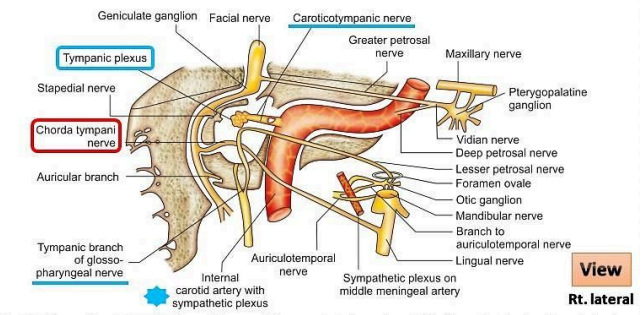


Fig. 25: Nerves in relation with the middle ear. Note secretomotor pathway of salivary, lacrimal and nasal glands

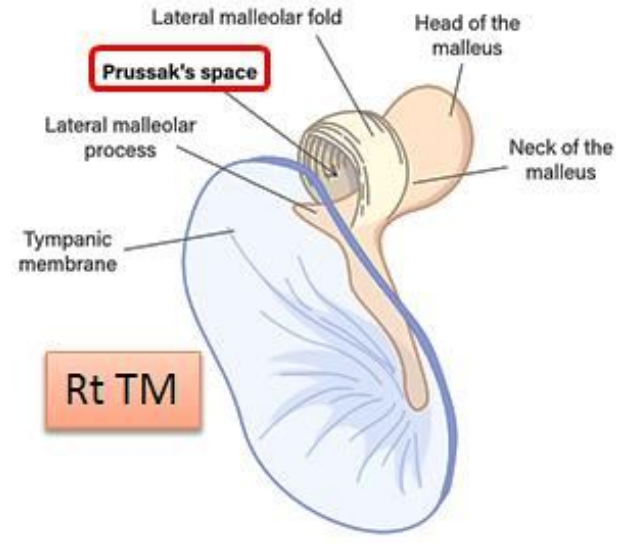
Nerves of the middle ear: tympanic plexus and petrosal nerves

ENT Steps Anatomy, p. 53

07 Mucosal folds and compartments

- Middle ear mucosa wraps the ossicles and ligaments like peritoneum; the folds divide the cleft into compartments that steer cholesteatoma.
- **Prussak's space:** lateral = pars flaccida; medial = neck of malleus; floor = lateral process of malleus; roof = lateral malleal ligament.
- Attic is divided by the superior malleal fold into a small anterior and a large posterior compartment.
- **Pouches of von Troeltsch:** anterior and posterior, between the malleal folds and the TM in front of and behind the malleus handle.
- Mucosa: anterior and inferior parts have ciliated cells and goblet cells (clearance); posterior and attic parts are flat cells (gas exchange). ET is lined by ciliated pseudostratified columnar epithelium.

> **CLINICAL PEARL** Pars flaccida cholesteatoma fills Prussak's space first, then spreads to the posterior attic and down the posterior pouch.



Prussak's space

ENT Steps Anatomy, p. 54

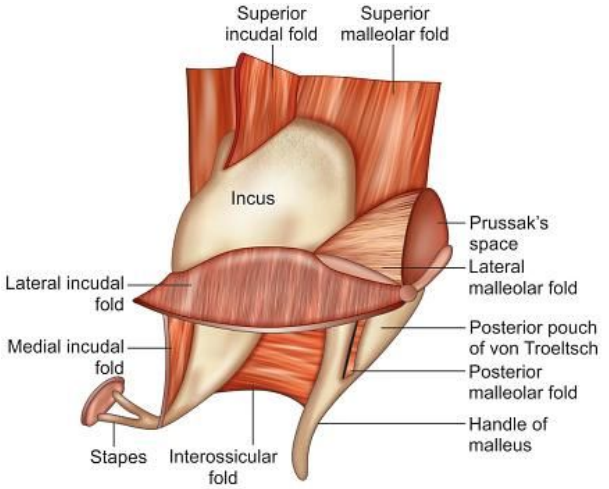


Fig. 26: Posterosuperior and lateral view of right tympanic cavity showing compartments and folds of middle ear (after Proctor)

Compartments and folds of the middle ear

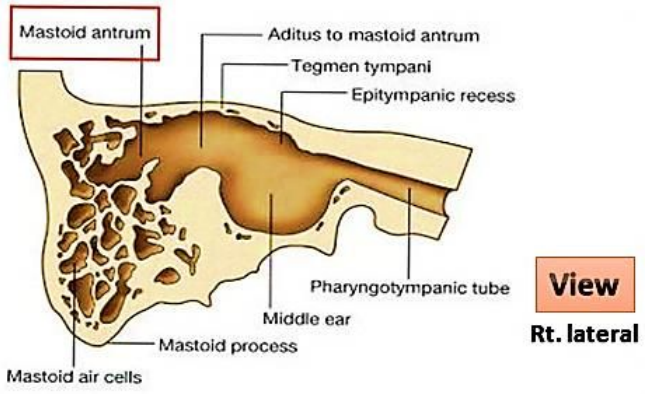
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08 Mastoid antrum and air cells

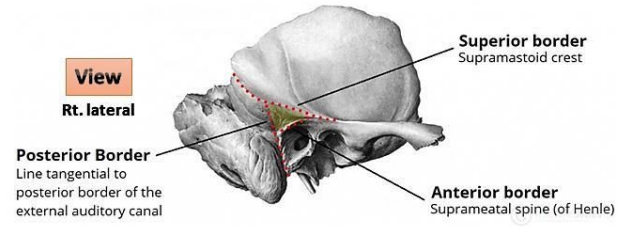
- **Antrum:** about 9 × 14 × 7 mm; present in all mastoids (most constant cell). Connected to the attic by the **aditus**.
- **Macewen's triangle** marks the antrum, about **1.5 cm** deep: above, the **temporal line**; front, posterosuperior canal margin with the **spine of Henle**; behind, a tangent to the back of the canal.

- **Relations:** roof = tegmen antri (middle fossa); medial = posterior SCC, endolymphatic sac, posterior fossa dura; behind = sigmoid sinus.
- **Mastoid types:** cellular (~80%), diploic, sclerotic (acellular). In a sclerotic mastoid the antrum is small and the sigmoid sinus may lie far forward.
- **Cell groups:** perilabyrinthine, perisinus, peritubal, tegmen, tip, zygomatic, retrofacial, squamous; peritubal and hypotympanic cells reach the petrous apex.
- **Körner's septum:** persistent petrosquamous plate that separates superficial squamous cells from deep petrous cells and can be mistaken for the medial wall of the antrum, so the antrum and deep cells are missed.

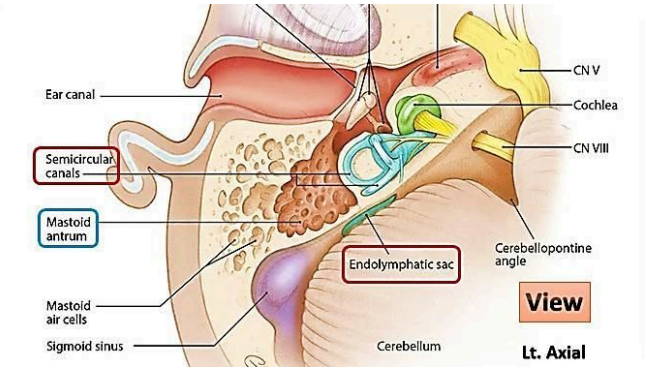
> **CLINICAL PEARL** Opening only the superficial cells above Körner's septum and missing the antrum is a classic cause of persistent mastoiditis after surgery.



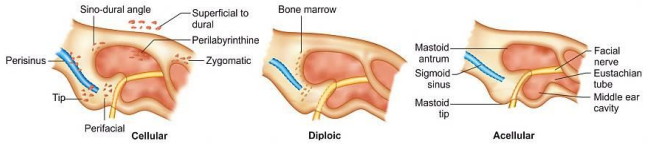
Mastoid antrum, aditus and epitympanic recess
ENT Steps Anatomy, p. 56



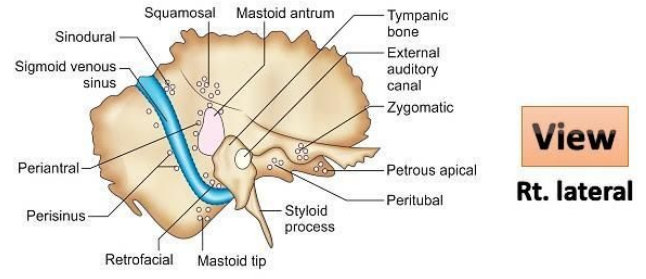
Macewen's (suprameatal) triangle
ENT Steps Anatomy, p. 56



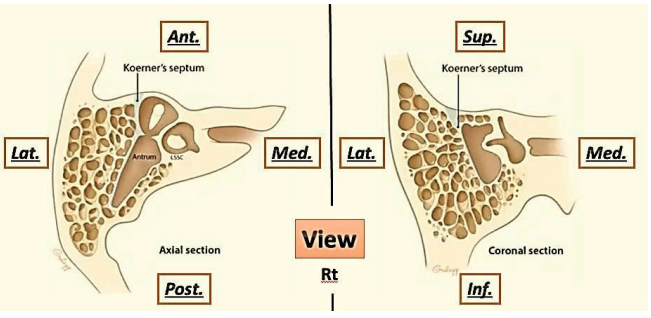
Axial relations of the middle ear and mastoid
ENT Steps Anatomy, p. 57



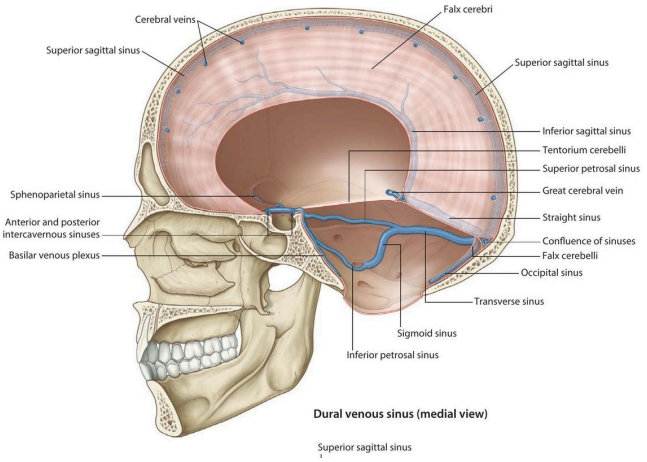
Types of mastoid: cellular, diploic, acellular
ENT Steps Anatomy, p. 59



Groups of mastoid air cells
ENT Steps Anatomy, p. 60

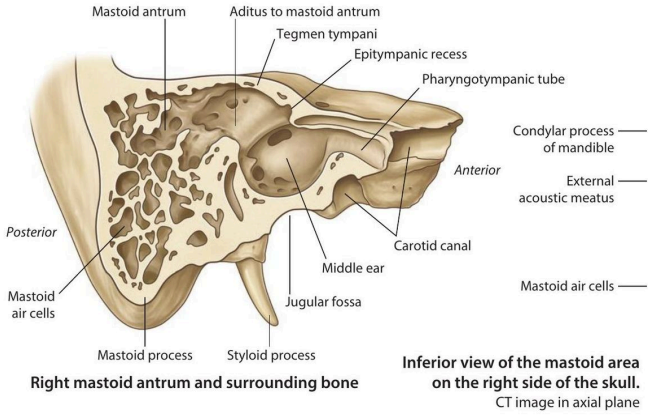


Korner's septum
ENT Steps Anatomy, p. 61



Dural venous sinuses (medial view): transverse, sigmoid and petrosal sinuses

Gray's Atlas of Anatomy, 2nd ed., p. 493



Right mastoid antrum and surrounding bone

Gray's Atlas of Anatomy, 2nd ed., p. 523

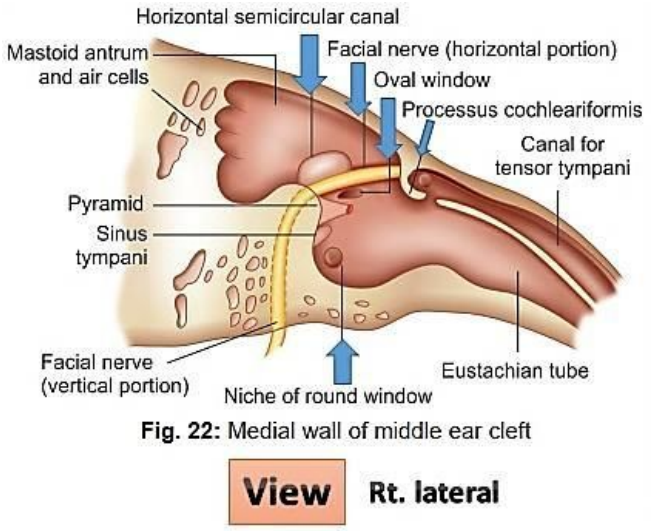
09 Blood supply and lymphatics

- Mainly from the ECA: **anterior tympanic** (maxillary artery) and **stylomastoid** (posterior auricular) are the major contributors; also superior tympanic and petrosal branches (middle meningeal), tympanic branch of ascending pharyngeal, artery of the pterygoid canal.
- ICA contributes caroticotympanic branches.
- Veins drain to the pterygoid plexus, superior petrosal sinus and sigmoid sinus.
- Lymph: middle ear → **retropharyngeal** and parotid nodes, then upper deep cervical; ET → retropharyngeal.

10 Vascular and nerve variants in the middle ear

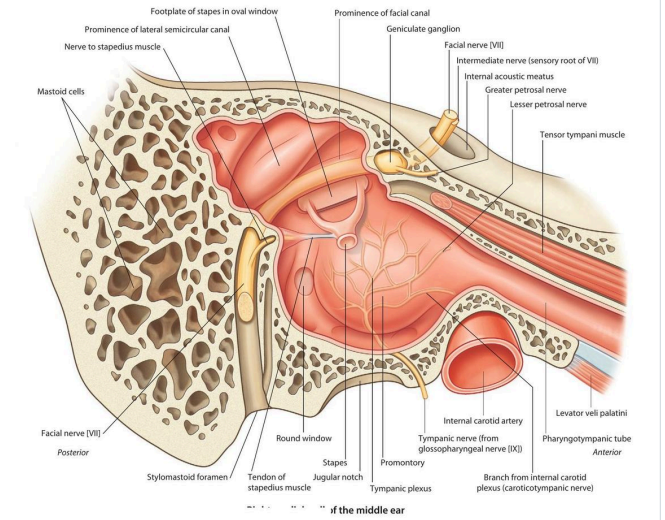
- **Dehiscent facial canal:** common, especially in the tympanic segment above the oval window. The nerve may bulge over the stapes or the oval window niche: a hazard in stapes and cholesteatoma surgery.
- **Displaced facial nerve:** in congenital ear malformations and aural atresia, the mastoid segment tends to lie more anterior and lateral. The nerve may also split or run below the oval window.
- **Aberrant internal carotid artery:** the ICA enters the middle ear through an enlarged inferior tympanic canaliculus and crosses the promontory, appearing as a **red retrotympanic mass**. On CT the vertical segment of the carotid canal is missing. Biopsy can cause catastrophic bleeding.
- **Persistent stapedial artery:** an embryonic branch of the ICA running through the stapes crura to the middle cranial fossa, where it replaces the middle meningeal artery. On CT the **foramen spinosum is absent**. It complicates stapes surgery.
- **High or dehiscent jugular bulb:** the bulb can rise above the floor of the middle ear or the annulus and may be uncovered, showing as a **blue retrotympanic mass**. It is at risk in myringotomy, tympanoplasty and cochlear implantation.
- **Anterior sigmoid sinus and low tegmen:** narrow the mastoid approach and the route to the antrum. Recognise them on the preoperative CT.

> **CLINICAL PEARL** A pulsatile red or blue mass behind the drum needs a CT before any biopsy: an aberrant carotid, a dehiscent jugular bulb and a paraganglioma can look alike.



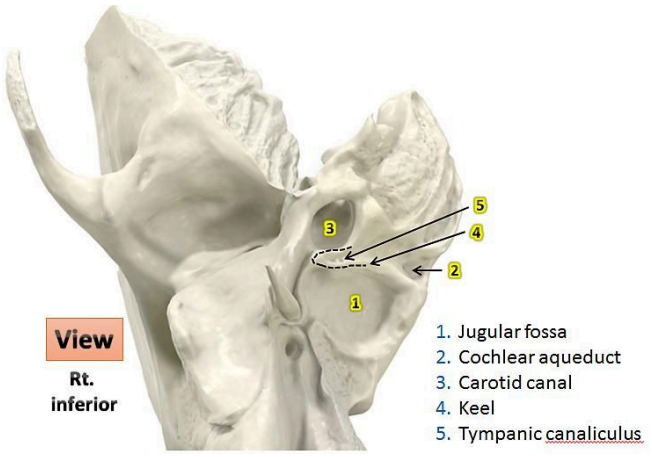
Medial wall of the middle ear cleft

ENT Steps Anatomy, p. 48



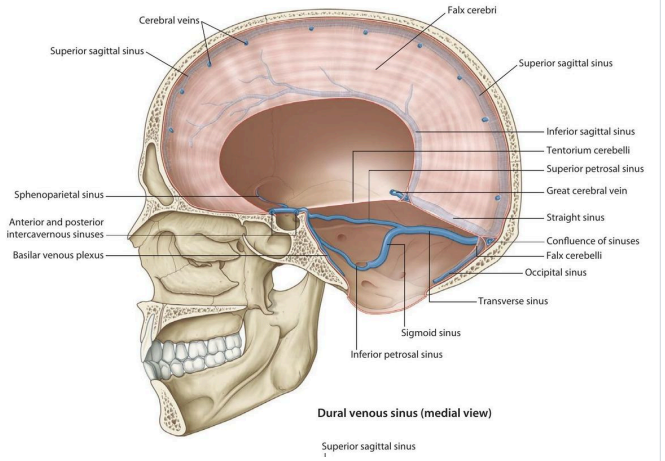
Right medial wall of the middle ear

Gray's Atlas of Anatomy, 2nd ed., p. 522



Inferior surface: jugular fossa, carotid canal, cochlear aqueduct

ENT Steps Anatomy, p. 24



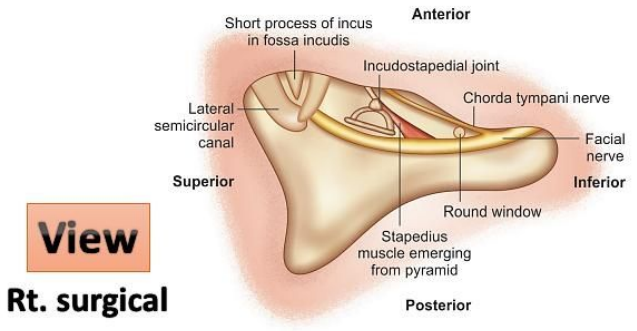
Dural venous sinuses (medial view): transverse, sigmoid and petrosal sinuses

Gray's Atlas of Anatomy, 2nd ed., p. 493

11 Surgical anatomy: In theatre

- **Cortical mastoidectomy:** drill within Macewen's triangle; the antrum lies about 1.5 cm deep. Define the tegmen above, sigmoid sinus behind, and the **lateral canal prominence** in the antrum, which marks the level of the facial nerve's second genu.
- **Facial nerve landmarks in the mastoid:** lateral canal, **short process of incus** (points to the facial recess), digastric ridge (points to the stylomastoid foramen).
- **Canal wall up vs canal wall down:** keeping the wall gives a normal canal but higher residual disease; taking it down exteriorises the attic and mastoid, needs meatoplasty and lifelong cleaning.
- **Posterior tympanotomy:** open the facial recess between the chorda tympani and vertical facial nerve, below the fossa incudis.
- **Stapes surgery:** the facial nerve lies above the oval window and may overhang it; the saccule lies about 1–2 mm below the footplate.

> **CLINICAL PEARL** Drill parallel to the facial nerve with a large diamond burr and plenty of irrigation; never drill across its course.

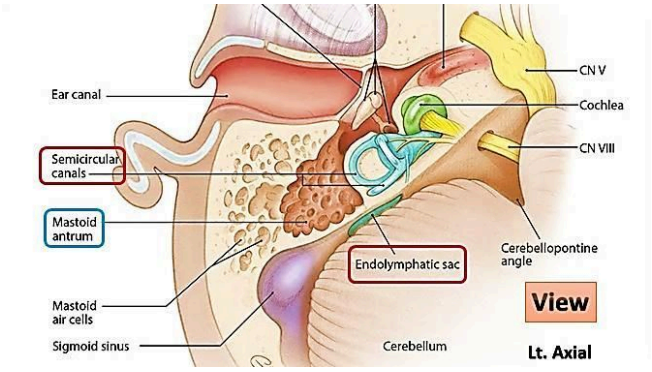


Posterior tympanotomy: middle ear seen through the facial recess

ENT Steps Anatomy, p. 46

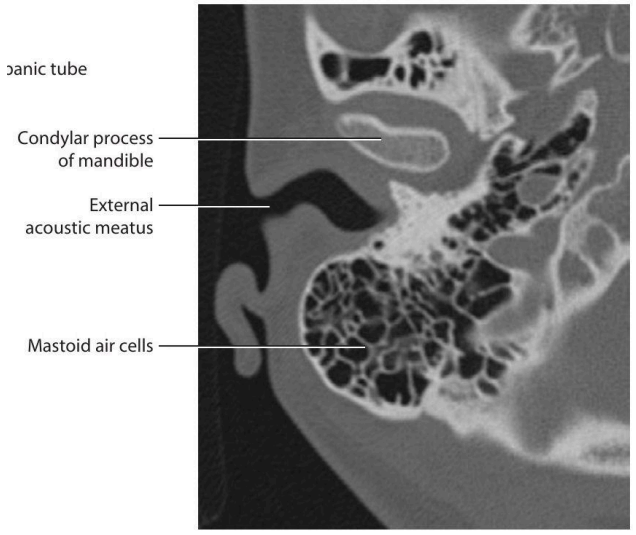
12 Radiological anatomy: On imaging

- **HRCT** shows ossicles, scutum erosion (early attic cholesteatoma), Prussak's space, tegmen and lateral canal fistula.
- **Non-EPI diffusion-weighted MRI** shows cholesteatoma as a bright spot; used to find residual or recurrent disease instead of second-look surgery.
- Opacified middle ear on CT does not distinguish fluid, granulation and cholesteatoma; bone erosion and the DWI signal do.



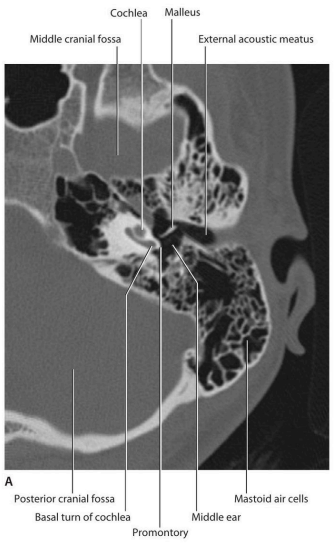
Axial relations of the middle ear and mastoid

ENT Steps Anatomy, p. 57



Axial CT of the mastoid air cells

Gray's Atlas of Anatomy, 2nd ed., p. 523



Axial CT through the ear (A): cochlea, malleus, mastoid

Gray's Atlas of Anatomy, 2nd ed., p. 525

Tables

Sensory supply of the ear

Area	Nerve	Origin
Tragus, front of auricle, anterosuperior EAC, anterior lateral TM	Auriculotemporal	V3
Concha, posteroinferior EAC, posterior lateral TM	Arnold's (auricular branch)	X
Concha and posterior canal wall (small area)	Sensory twigs of the facial nerve	VII
Most of the medial auricle, lobule, back of lateral surface	Great auricular	C2, C3
Upper medial auricle and postauricular skin	Lesser occipital	C2
Medial surface of TM, middle ear, mastoid, bony ET	Jacobson's (tympanic plexus)	IX

Referred otalgia comes through V3, VII, IX, X and C2–C3. Throat cancer reaches the ear through IX and X.

Middle ear muscles

	Tensor tympani	Stapedius
Arch	1st	2nd
Origin	Canal above the ET	Pyramid
Insertion	Neck of malleus (round the processus cochleariformis)	Neck of stapes
Nerve	V3 (nerve to medial pterygoid)	Nerve to stapedius (VII)
Action	Tenses the TM, pulls malleus inward	Dampens stapes movement (acoustic reflex)

Types of mastoid surgery

Operation	What is removed	What is kept	Main use
Cortical (simple) mastoidectomy	Mastoid air cells and antrum	Canal wall, middle ear	Acute mastoiditis, access for CI
Canal wall up (combined approach)	Mastoid cells, posterior tympanotomy	Posterior canal wall	Cholesteatoma with good ET, children

Operation	What is removed	What is kept	Main use
Modified radical (canal wall down)	Posterior canal wall, attic lateral wall, disease	Middle ear and ossicles where possible, TM	Extensive cholesteatoma
Radical mastoidectomy	Canal wall, TM remnant, ossicles except stapes; middle ear and mastoid exteriorised; ET closed	Stapes (if present)	Unresectable disease, poor hearing ear
Atticotomy	Scutum and attic disease	Rest of canal wall	Limited attic cholesteatoma

Pars flaccida vs pars tensa cholesteatoma

	Attic (pars flaccida)	Posterior mesotympanic (pars tensa)
Origin	Retraction of Shrapnell's membrane	Posterosuperior pars tensa retraction
First space	Prussak's space	Posterior pouch, sinus tympani, facial recess
Spread	Posterior attic → aditus → antrum	Sinus tympani, facial recess, under the incus
Ossicle often eroded	Head of malleus, body of incus	Long process of incus, stapes superstructure

Tympanogram types

Type	Pattern	Typical cause
A	Normal peak at atmospheric pressure	Normal middle ear
As	Shallow peak	Otosclerosis, tympanosclerosis
Ad	Very tall peak	Ossicular discontinuity, flaccid TM
B	Flat	Effusion (normal volume), perforation or patent grommet (high volume), wax (low volume)
C	Peak at negative pressure	Eustachian tube dysfunction

Tuning fork tests

Finding	Rinne (affected ear)	Weber
Normal	Positive (AC > BC)	Central
Conductive loss	Negative (BC > AC) when the air-bone gap is about 25–30 dB or more with 512 Hz (15–20 dB with 256 Hz)	Lateralises to the affected ear
Sensorineural loss	Positive (both reduced)	Lateralises to the better ear
Dead ear	False negative (sound heard in the other ear by bone)	To the better ear

Use a 512 Hz fork.

Development of the ear

Part	Origin	Timing / notes
Inner ear	Otic placode → otic vesicle	Week 4; adult size by ~week 20–25
Middle ear and ET	1st pouch (tubotympanic recess)	Pneumatisation continues after birth
Malleus head, incus body	1st arch	—

Part	Origin	Timing / notes
Handle of malleus, long process of incus, stapes superstructure	2nd arch	Footplate partly otic capsule
EAC	1st cleft (meatal plug canalises)	Weeks 21–28
Auricle	Six hillocks of His (1st and 2nd arches)	Weeks 6–8
Mastoid process	Grows after birth	Absent at birth; facial nerve superficial

M Mnemonics

Medial wall, top to bottom

Canal, Nerve, Oval, Promontory, Round

Lateral canal prominence, facial nerve, oval window, promontory, round window

Middle ear contents

3-2-2

3 ossicles, 2 muscles, 2 nerves (chorda tympani, tympanic plexus)

Macewen's triangle

Line, Henle, tangent

Temporal line above, spine of Henle in front, tangent to the posterior canal behind

Prussak's space walls

Flaccid, Neck, Process, Ligament

Lateral pars flaccida, medial malleus neck, floor lateral process, roof lateral malleal ligament

Q Rapid recall

1. **What lies on the medial wall of the middle ear?** Promontory, oval window, round window, horizontal facial canal, lateral SCC prominence, processus cochleariformis.
2. **Boundaries of the facial recess?** Medial: vertical facial nerve. Lateral: chorda tympani and annulus. Superior: fossa incudis.
3. **Why is residual cholesteatoma common in the sinus tympani?** It lies medial to the facial nerve and pyramid and is hidden from both the canal and mastoid views.
4. **Nerve supply and action of tensor tympani and stapedius?** Tensor tympani: V3, tenses the TM. Stapedius: nerve to stapedius (VII), dampens stapes movement.
5. **Route of the chorda tympani in the middle ear?** Enters through the posterior canaliculus, crosses the medial TM between the malleus handle and long process of incus, leaves through the petrotympanic fissure.
6. **Boundaries of Prussak's space?** Lateral: pars flaccida. Medial: neck of malleus. Floor: lateral process of malleus. Roof: lateral malleal ligament.
7. **Boundaries of Macewen's triangle?** Temporal line above, posterosuperior canal margin (spine of Henle) in front, tangent to the posterior canal wall behind. Antrum lies about 1.5 cm deep.
8. **What is Körner's septum?** Persistent petrosquamous plate separating superficial squamous from deep petrous mastoid cells; can hide the antrum.
9. **Main arteries of the middle ear?** Anterior tympanic (maxillary) and stylomastoid (posterior auricular).
10. **Secretomotor pathway to the parotid?** Inferior salivatory nucleus → IX → Jacobson's nerve → tympanic plexus → lesser petrosal → otic ganglion → auriculotemporal → parotid.
11. **Volume of the tympanic cavity?** About 1–2 cm^3 (the whole cleft is larger, depending on mastoid pneumatisation).
12. **Which part of the middle ear lies above the pars tensa?** Epitympanum (attic).
13. **What is the scutum?** The bony lateral wall of the attic, above the TM; eroded early in attic cholesteatoma.
14. **What forms the floor of the middle ear?** A thin plate of bone over the jugular bulb (may be dehiscent).
15. **What lies in the anterior wall of the middle ear?** ET opening, canal for tensor tympani above it, anterior canaliculus for chorda tympani, and the carotid canal behind thin bone.

16. **What is the processus cochleariformis?** Hook of bone in front of the oval window around which the tensor tympani tendon turns; a landmark for the facial nerve.
17. **Size of the malleus and stapes?** Malleus about 8 mm (largest); stapes about 3.5 mm (smallest bone in the body).
18. **Which arch forms the stapes superstructure?** The 2nd arch.
19. **Pressure gain of the middle ear transformer?** About 22 times (area ratio ~17:1 × lever ~1.3:1), roughly 25–30 dB.
20. **Where does the chorda tympani leave the middle ear?** Through the petrotympanic (Glaserian) fissure.
21. **What is the commonest mastoid type?** Cellular (well pneumatised), about 80%.
22. **Which mastoid cells connect to the petrous apex?** Peritubal and hypotympanic cells, and perilabyrinthine cells.
23. **What is the depth of the antrum below Macewen's triangle?** About 1.5 cm in adults.
24. **What does the lateral canal prominence mark during mastoidectomy?** The level of the facial nerve's second genu, just below and in front of it.
25. **Two compartments of the attic?** A small anterior and a large posterior compartment, separated by the superior malleal fold.
26. **CT sign of a persistent stapedial artery?** Absent foramen spinosum (it replaces the middle meningeal artery).
27. **CT sign of an aberrant internal carotid artery in the middle ear?** Missing vertical segment of the carotid canal, with the vessel entering through an enlarged inferior tympanic canaliculus.
28. **Blue mass behind an intact drum, not pulsatile: most likely vascular cause?** A high, dehiscent jugular bulb.

Figures from ENT Steps Anatomy (based on Bansal) and Dr. Sameh Doss's Anatomy of Head & Neck handout, for personal study.